1 The Jordan–Wigner mapping for spinless Fermions

1.1 Summary

The Jordan–Wigner transformation (JWT) can be used to map a fermionic system to a system of spins. This is essentially done by rewriting

$$\text{fermion} = \text{string} \times \text{operator},$$

where the "string" is some product of operators $\hat{Z}_i = 1 - 2\hat{n}_i$ and the "operator" is found to obey spin commutation relations.

1.2 Setup: the fermionic Hilbert space

Our fermionic Hilbert space is defined by (Nielsen):

i) operators $c_1, \ldots, c_L$ acting on some Hilbert space;

ii) these operators satisfying the canonical (anti-)commutation relations for fermions

$$\{c_i^{(+)}, c_j^{(+)}\} = 0 \qquad \& \qquad \{c_i, c_j^\dagger\} = \delta_{ij}\, \mathbb{1}. \tag{1}$$

Many things directly follow from this, such as a possible basis for the Hilbert space, namely the occupation number basis

$$|\boldsymbol{n}\rangle = |n_1, n_2, \ldots, n_L\rangle \equiv \left(\prod_{i=1}^{L} (c_i^\dagger)^{n_i}\right) |0\rangle\,, \tag{2}$$

where $\hat{n}_i\,|\boldsymbol{n}\rangle = c_i^\dagger c_i\,|\boldsymbol{n}\rangle = n_i\,|\boldsymbol{n}\rangle$ with $n_i = 0, 1$ and

$$[\hat{n}_i, \hat{n}_j] = 0. \tag{3}$$

1.3 The transformation

The JWT introduces new operators a_j by writing

$$c_j \equiv \left(\prod_{i=1}^{j-1} Z_i\right) a_j \qquad \& \qquad c_j^\dagger \equiv a_j^\dagger \left(\prod_{i=1}^{j-1} Z_i\right), \tag{4}$$

with $Z_j = (-1)^{n_j} = 1 - 2n_j$ and $n_j = c_j^\dagger c_j$. The ordering of the Z_i does not matter since $[Z_i, Z_j] = 0$ due to eq. (3). Inverting eq. (4) using $Z_i^2 = \mathbb{1}$ and $[Z_i, Z_j] = 0$ yields

$$a_j = \left(\prod_{i=1}^{j-1} Z_i\right) c_j \qquad \& \qquad a_j^\dagger = c_j^\dagger \left(\prod_{i=1}^{j-1} Z_i\right). \tag{5}$$

1.4 Algebra of the new operators

One can show that these new operators obey

$$[a_i^{(+)}, a_j^{(+)}] = 0 \qquad \forall\, i, j, \tag{6}$$

$$[a_i, a_j^\dagger] = 0 \qquad \forall\, i \neq j, \tag{7}$$

$$[a_i, a_i^\dagger] = 1 - 2n_i \qquad \text{with } n_i = c_i^\dagger c_i, \tag{8}$$

using that

$$\{Z_i, c_j^{(+)}\} = 0 \tag{9}$$

The corresponding proofs can be found in section 1.8.

1.5 The occupation-number basis in the new operators

The basis states of the fermionic Hilbert space can be written as

$$\begin{aligned}
|n_1, n_2, \ldots, n_L\rangle &\equiv \left(\prod_{i=1}^{L} (c_i^\dagger)^{n_i}\right) |0\rangle && \text{w. } n_i = 0, 1 \\
&= \left\{\prod_{i=1}^{L} \left[a_i^\dagger \left(\prod_{j=1}^{i-1} Z_j\right)\right]^{n_i}\right\} |0\rangle && (c_1^\dagger = a_1^\dagger) \\
&= (a_1^\dagger)^{n_1} (a_2^\dagger Z_1)^{n_2} (a_3^\dagger Z_2 Z_1)^{n_3} \cdots (a_L^\dagger Z_{L-1} \cdots Z_1)^{n_L} |0\rangle \\
&= \left(\prod_{i=1}^{L} (a_i^\dagger)^{n_i}\right) (Z_1)^{n_2} (Z_2 Z_1)^{n_3} \cdots (Z_{L-1} \cdots Z_1)^{n_L} |0\rangle \\
&= \left(\prod_{i=1}^{L} (a_i^\dagger)^{n_i}\right) |0\rangle && \text{w. } n_i = 0, 1,
\end{aligned} \tag{10}$$

using $[Z_i, Z_j] = 0\ \forall i, j$, $[Z_i, c_j] = 0$ for $i \neq j$ (hence $[Z_i, a_j^\dagger] = 0$ for $i \neq j$), and $Z_j |0\rangle = |0\rangle$. The ordering is not important anymore because $[a_i^\dagger, a_j^\dagger] = 0$.

1.6 Action on basis states

One cannot define the action of a_j ($a_j^\dagger$) locally, since the JW string $Z_1 \cdots Z_{j-1}$ is non-local, so one has to examine them on the full many-body basis. For $n_\ell = 0$,

$$\begin{aligned}
a_\ell |n_1, \ldots, n_\ell = 0, \ldots, n_L\rangle &= a_\ell \left(\prod_{i=1}^{L} (a_i^\dagger)^{n_i}\right) |0\rangle \\
&\overset{[a_\ell, a_j^\dagger]=0}{=} \left(\prod_{i=1}^{L} (a_i^\dagger)^{n_i}\right) a_\ell |0\rangle = 0,
\end{aligned} \tag{11}$$

since $a_\ell |0\rangle = c_\ell |0\rangle = 0$. For $n_\ell = 1$,

$$\begin{aligned} a_\ell |n_1, \ldots, n_\ell = 1, \ldots, n_L\rangle &= a_\ell \left(\prod_{\substack{i=1 \\ i\neq\ell}}^{L} (a_i^\dagger)^{n_i} \right) a_\ell^\dagger |0\rangle \\ &\overset{[a_\ell, a_j^\dagger]=0}{=} \left(\prod_{\substack{i=1 \\ i\neq\ell}}^{L} (a_i^\dagger)^{n_i} \right) a_\ell a_\ell^\dagger |0\rangle = \left(\prod_{\substack{i=1 \\ i\neq\ell}}^{L} (a_i^\dagger)^{n_i} \right) c_\ell c_\ell^\dagger |0\rangle \\ &= \left(\prod_{\substack{i=1 \\ i\neq\ell}}^{L} (a_i^\dagger)^{n_i} \right) |0\rangle = |n_1, \ldots, n_\ell = 0, \ldots, n_L\rangle . \end{aligned} \tag{12}$$

Using $[a_\ell^\dagger, a_j^\dagger] = 0$ and $(a_\ell^\dagger)^2 = 0$, one quickly finds

$$a_\ell^\dagger |n_1, \ldots, n_\ell = 0, \ldots, n_L\rangle = |n_1, \ldots, n_\ell = 1, \ldots, n_L\rangle , \tag{13}$$

$$a_\ell^\dagger |n_1, \ldots, n_\ell = 1, \ldots, n_L\rangle = 0. \tag{14}$$

Thus $a_\ell\,(a_\ell^\dagger)$ destroys (creates) a particle in the many-body state *without* giving rise to an extra sign structure (as $c_\ell\,(c_\ell^\dagger)$ would do).

1.7 Identification with spins

We can thus identify $n_i = 0, 1$ with $n_i = \downarrow, \uparrow$ in the many-body state, and the new operators can be associated with spins by identifying

$$a_j = S_j^-, \qquad a_j^\dagger = S_j^+ \qquad \& \qquad n_j - \tfrac{1}{2} = S_j^z = \tfrac{1}{2}\,\sigma_j^z, \tag{15}$$

since $[S_j^-, S_j^+] = -2S_j^z$ for spins (see the quick refresher on spin, **??**). For instance

$$a_3^\dagger |0\rangle = S_3^+ |\downarrow\downarrow\downarrow\rangle = |\uparrow\downarrow\downarrow\rangle .$$

Matrix elements do not change under the JWT, since it is only a rewriting of operators and a renaming of basis states.

1.8 Proofs about the JWT

Claim: $\{Z_j, c_j^{(+)}\} = 0$**.**

Proof. On the single site j, with $Z_j |0\rangle_j = |0\rangle_j$, $Z_j |1\rangle_j = -|1\rangle_j$, $c_j |0\rangle_j = 0$ and $c_j |1\rangle_j = |0\rangle_j$,

$$Z_j c_j |0\rangle_j = Z_j \cdot 0 = 0 = c_j |0\rangle_j = c_j Z_j |0\rangle_j , \tag{16}$$

$$Z_j c_j |1\rangle_j = Z_j |0\rangle_j = |0\rangle_j = c_j |1\rangle_j , \tag{17}$$

$$c_j Z_j |1\rangle_j = -c_j |1\rangle_j = -|0\rangle_j . \tag{18}$$

Hence, on both basis states $\{Z_j, c_j\} = Z_j c_j + c_j Z_j = c_j - c_j = 0$, and since $Z_j^\dagger = Z_j$ it follows that $\{Z_j, c_j^\dagger\} = 0$ as well. (see Coleman, Chapter 4.1) □

Claim: $[a_i^{(+)}, a_j^{(+)}] = 0 \;\; \forall\, i, j$**.**

Proof. Assume w.l.o.g. that $i < j$. Using eq. (5),

$$a_j a_i = (Z_1 \cdots Z_{j-1} c_j)\,(Z_1 \cdots Z_{i-1} c_i) \overset{\substack{[Z_j, c_\ell]=0,\, j\neq\ell \\ \{c_i, c_j\}=0}}{=} Z_1^2 \cdots Z_{i-1}^2\, Z_i \cdots Z_{j-1}\, c_j c_i = -Z_i \cdots Z_{j-1}\, c_i c_j \qquad (= 0 \text{ if } i = j), \tag{19}$$

and likewise

$$a_i a_j = (Z_1 \cdots Z_{i-1} c_i)\,(Z_1 \cdots Z_{j-1} c_j) \overset{[Z_i, c_\ell]=0,\, i\neq\ell}{=} Z_{i+1} \cdots Z_{j-1}\, c_i Z_i c_j \overset{\{Z_i, c_i\}=0}{=} -Z_i \cdots Z_{j-1}\, c_i c_j. \tag{20}$$

Comparing the two, $a_i a_j = a_j a_i$, i.e. $[a_i, a_j] = 0$. Analogously for $[a_i^\dagger, a_j^\dagger]$ and for $i > j$; the case $i = j$ is straightforward since $c_i^2 = 0$. □

Claim: $[a_i, a_j^\dagger] = 0 \;\; \forall\, i \neq j$**.**

Proof. We can show it w.l.o.g. for $i < j$. Using eq. (5) and eq. (9),

$$\begin{aligned} a_i^\dagger a_j &= \big(c_i^\dagger Z_{i-1} \cdots Z_1\big)\big(Z_1 \cdots Z_{j-1} c_j\big) \overset{Z_i^2=1,\, i<j}{=} c_i^\dagger\, Z_i \cdots Z_{j-1}\, c_j \\ &= Z_{i+1} \cdots Z_{j-1}\, c_i^\dagger Z_i c_j \overset{\{Z_i, c_i^\dagger\}=0}{=} -Z_i \cdots Z_{j-1}\, c_i^\dagger c_j = +Z_i \cdots Z_{j-1}\, c_j c_i^\dagger, \end{aligned} \tag{21}$$

and

$$\begin{aligned} a_j a_i^\dagger &= \big(Z_1 \cdots Z_{j-1} c_j\big)\big(c_i^\dagger Z_{i-1} \cdots Z_1\big) \\ &= Z_i \cdots Z_{j-1}\, c_j c_i^\dagger\, Z_{i-1}^2 \cdots Z_1^2 \qquad ([Z_j, c_\ell^{(+)}] = 0,\; j \neq \ell) \\ &= Z_i \cdots Z_{j-1}\, c_j c_i^\dagger. \end{aligned} \tag{22}$$

Both reduce to the same expression, so $a_j a_i^\dagger - a_i^\dagger a_j = 0$ for $i < j$, i.e. $[a_j, a_i^\dagger] = 0$; the case $i > j$ follows straightforwardly. □

Claim: $[a_i, a_i^\dagger] = 1 - 2n_i$ **with** $n_i = c_i^\dagger c_i$**.**

Proof. Using eq. (5) and $Z_k^2 = \mathbb{1}$,

$$a_i^\dagger a_i = c_i^\dagger\, Z_{i-1} \cdots Z_1 Z_1 \cdots Z_{i-1}\, c_i = c_i^\dagger c_i = n_i, \tag{23}$$

$$a_i a_i^\dagger = Z_1 \cdots Z_{i-1}\, c_i c_i^\dagger\, Z_{i-1} \cdots Z_1 = c_i c_i^\dagger = 1 - c_i^\dagger c_i, \tag{24}$$

so that

$$[\hat{a}_i, \hat{a}_i^\dagger] = a_i a_i^\dagger - a_i^\dagger a_i = (1 - c_i^\dagger c_i) - c_i^\dagger c_i = 1 - 2\hat{n}_i. \tag{25}$$

□

1.9 References

- *Fermions and Jordan–Wigner String* – ITensor C++ Documentation,
- *Introduction to Many-Body Physics*, Chapter 4.1 – Piers Coleman,
- *The Fermionic canonical commutation relations and the Jordan–Wigner transform* – M. Nielsen,